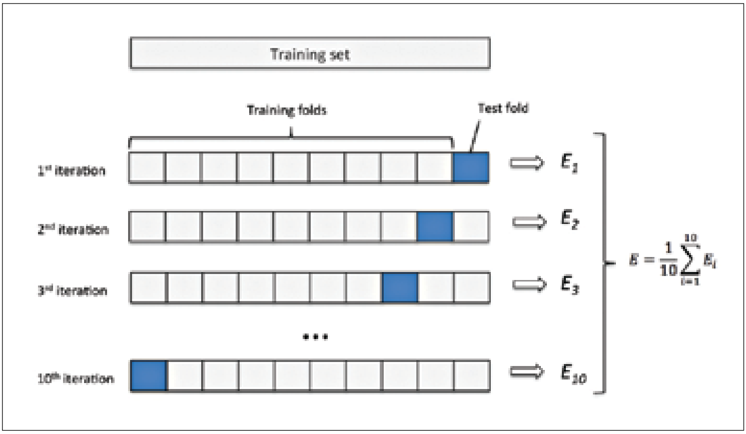
K-Fold Cross Validation

There is almost never enough data to accurately train a model. Thus removing any data for validating can cause a loss in accuracy by model underfitting. The model may fail to notice certain patterns or make false predictions based on the limited (maybe biased) dataset. So, we need a method that provides sufficient data for training the model and also leaves sufficient data for validation. K Fold cross validation is the answer to this problem.

In this method, the data is divided into k subsets. Now we repeat the holdout method k times, during each time, one of the k subsets is used as the validation set and the other k-1 subsets are combined to form a training set. The model efficiency is calculated by averaging over all k trials. Every data point is in a validation set exactly once and is in a training set k-1 times. This reduces bias as most of the data is used for fitting, and the variance is also remarkably reduced as each data set is used only once in validation. The value of K is usually taken to be five or ten. There is no standard rule to it.



K Fold cross validation

Stratified K-Fold Cross Validation

In few cases, there is an imbalance in the response variables due to an inherent bias in the nature of the dataset. We use a variation of the K Fold cross validation technique in such cases. To make sure that each fold contains approximately the same percentage of samples of each class as the actual dataset or the mean prediction percent of each class is proportional to the actual dataset. This is also known as Stratified K Fold Cross Validation.